DigiLocker Issuer — Database Design Document (v2.1 FINAL)

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1. Objective

Design a **PostgreSQL schema** that:

- Enforces **DLTS-compliant document identity (URI + doc_id)**
- Supports **lazy, atomic URI assignment**
- Guarantees **immutability of identifiers**
- Enables **secure lookup via (authorization_number + document_type)**
- Provides **auditability and integrity tracking**
- Remains **portable across environments**

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2. Core Design Principles

2.1 Identity Model

Lookup Key → (authorization_number, document_type)

Public Identifier → URI

Internal ID → doc_id (random, non-guessable)

2.2 URI Rules

- Format:
- <ISSUER_ID>-<DocType>-<doc_id>
- Generated **once**
- Never updated
- Globally unique

2.3 Responsibility Split

Concern	Layer
Uniqueness, immutability	DB
Identity validation (DOB/name)	Application
Authentication (HMAC)	Application

2.4 Storage Strategy

- Store **relative file paths only**
- Resolve at runtime using config

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3. Schema Overview

documents

access_logs

integrity_logs

(optional) document_types

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4. Core Table: documents

4.1 Final DDL

```
CREATE TABLE documents (
    id BIGSERIAL PRIMARY KEY,

    -- Business lookup
    authorization_number VARCHAR(50) NOT NULL,
    document_type VARCHAR(5) NOT NULL,
```

```
    -- DLTS identifiers (write-once)
    doc_id VARCHAR(10),
    uri VARCHAR(255),

    -- Identity attributes
    employee_name TEXT,
    employee_dob DATE,

    -- File metadata
    file_relative_path TEXT NOT NULL,
    file_checksum VARCHAR(64), -- SHA256 hex
    file_size_bytes BIGINT,
    file_last_checked_at TIMESTAMP,

    -- State
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    digilocker_enabled BOOLEAN NOT NULL DEFAULT TRUE,

    -- Audit
    created_at TIMESTAMP NOT NULL DEFAULT NOW(),
    updated_at TIMESTAMP NOT NULL DEFAULT NOW()
);
```

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5. Constraints

5.1 Business Uniqueness

```
ALTER TABLE documents
ADD CONSTRAINT uq_documents_business
UNIQUE (authorization_number, document_type);
```

5.2 DLTS Identity Constraints

```
ALTER TABLE documents
ADD CONSTRAINT uq_doc_id UNIQUE (doc_id);

ALTER TABLE documents
ADD CONSTRAINT uq_uri UNIQUE (uri);
```

5.3 Lifecycle Constraint (Critical)

```
ALTER TABLE documents
ADD CONSTRAINT chk_doc_id_uri_pair
CHECK (
   (doc_id IS NULL AND uri IS NULL)
   OR
   (doc_id IS NOT NULL AND uri IS NOT NULL)
);
```

5.4 Format Constraints

```
ALTER TABLE documents
ADD CONSTRAINT chk_document_type_format
CHECK (document_type ~ '^[A-Za-z]{1,5}$');

ALTER TABLE documents
```

```
ADD CONSTRAINT chk_doc_id_length
CHECK (doc_id IS NULL OR length(doc_id) <= 10);

ALTER TABLE documents
ADD CONSTRAINT chk_file_size
CHECK (file_size_bytes IS NULL OR file_size_bytes >= 0);
```

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6. Immutability Enforcement

6.1 Trigger

```
CREATE OR REPLACE FUNCTION prevent_identifier_update()
RETURNS trigger AS $$
BEGIN
   IF OLD.uri IS NOT NULL AND NEW.uri IS DISTINCT FROM OLD.uri THEN
      RAISE EXCEPTION 'URI is immutable';
   END IF;

   IF OLD.doc_id IS NOT NULL AND NEW.doc_id IS DISTINCT FROM OLD.doc_id THEN
      RAISE EXCEPTION 'doc_id is immutable';
   END IF;

   RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_prevent_identifier_update
```

```
BEFORE UPDATE ON documents
FOR EACH ROW
EXECUTE FUNCTION prevent_identifier_update();
```

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7. Index Strategy

7.1 Primary Lookup

```
CREATE INDEX idx_documents_lookup
ON documents (authorization_number, document_type)
WHERE is_active = TRUE AND digilocker_enabled = TRUE;
```

7.2 Identifier Access

```
CREATE INDEX idx_documents_uri ON documents (uri);
CREATE INDEX idx_documents_doc_id ON documents (doc_id);
```

7.3 Optional

```
CREATE INDEX idx_documents_created ON documents (created_at);
```

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8. File Handling Model

8.1 Storage

file_relative_path → stored

BASE_STORAGE_PATH → config

8.2 Runtime Resolution

full_path = BASE_STORAGE_PATH + file_relative_path

8.3 Integrity

- SHA256 checksum
- periodic verification
- logged discrepancies

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9. Access Logs

9.1 Table

```
CREATE TABLE access_logs (
    id BIGSERIAL PRIMARY KEY,

    document_id BIGINT REFERENCES documents(id) ON DELETE SET NULL,

    authorization_number VARCHAR(50),
    document_type VARCHAR(5),

    txn_id VARCHAR(100),
    digilocker_id VARCHAR(255),

    request_ip INET,
    user_agent TEXT,

    response_status SMALLINT,
    error_message TEXT,
```

```
    processing_time_ms INTEGER,

    created_at TIMESTAMP DEFAULT NOW()
);
```

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10. Integrity Logs

10.1 Table

```
CREATE TABLE integrity_logs (
    id BIGSERIAL PRIMARY KEY,

    document_id BIGINT REFERENCES documents(id) ON DELETE SET NULL,

    issue_type VARCHAR(30),
    stored_checksum VARCHAR(64),
    calculated_checksum VARCHAR(64),

    file_path TEXT,
    action_taken VARCHAR(20),

    created_at TIMESTAMP DEFAULT NOW()
);
```

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11. Concurrency Model

11.1 Requirement

- URI must be generated **exactly once**

11.2 Mechanism

Application must use:

SELECT * FROM documents WHERE id = ? FOR UPDATE;

11.3 Guarantees

- No duplicate URI
- No race conditions
- Atomic assignment

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12. Migration from Current Schema

Your current DB includes:

- string dates
- absolute file paths
- strict enum constraints

12.1 Migration Plan

Step 1 — Add new DATE columns

ALTER TABLE digilocker_documents ADD COLUMN employee_dob_new DATE;

ALTER TABLE digilocker_documents ADD COLUMN authorization_date_new DATE;

Step 2 — Backfill

UPDATE digilocker_documents

```
SET employee_dob_new = TO_DATE(employee_dob, 'DD/MM/YYYY'),
    authorization_date_new = TO_DATE(authorization_date, 'DD/MM/YYYY');
```

Step 3 — Replace old columns

- drop VARCHAR columns
- rename new columns

Step 4 — Convert file path

```
ALTER TABLE digilocker_documents RENAME COLUMN file_path TO file_relative_path;
```

Remove absolute path validation trigger

Step 5 — Relax document_type constraint

Drop:

chk_document_type

Step 6 — Add immutability trigger

Step 7 — Add lifecycle constraint

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13. What DB Does NOT Handle

- ❌ HMAC authentication
- ❌ identity validation (DOB/name)
- ❌ business rules
- ❌ DigiLocker request validation

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14. Security Considerations

- restrict DB access
- encrypt backups
- monitor audit logs
- no secrets stored in DB

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15. Performance Characteristics

Aspect	Behavior
Lookup	indexed, O(log n)
Writes	low volume
Locks	row-level only
Scaling	horizontal read possible

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16. Acceptance Criteria

DB is production-ready if:

- ✓ URI/doc_id immutable
- ✓ no duplicate URI possible
- ✓ safe concurrent generation
- ✓ lookup fast
- ✓ portable storage paths
- ✓ correct data types

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17. Final Alignment Status

Area	Status
Spec compliance	✓
Your original DB	✓ (with fixes)
Security	✓
Concurrency	✓
Maintainability	✓